

X-Wing: Propeller-Based Force Feedback to Head in a Virtual Environment

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ABSTRACT

In this demonstration, we present X-Wing, a force feedback device using ducted fans attached to a Virtual Reality (VR) Head Mounted Display (HMD). Applying haptic technologies to VR has become a standard solution to enhance the immersive experience. It is also essential for virtual interactions where user experience and performance are crucial. This research focuses on a head-based, wearable device that provides forces to the VR user using exertion force by electric thrusters with controllable strengths and directions. Our system allows the user to experience forces based on their virtual momentum and velocity. It enables a VR user to receive force feedback based on different thrust power converted into translational and rotational force for a unique VR experience such as virtual aerial simulation. We present the prototype to the attendees of SIGGRAPH Asia 2021 through a live demonstration.

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*; **Haptic devices**.

KEYWORDS

haptic feedback, virtual reality, head-mounted display, fan-based feedback

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1 INTRODUCTION

Haptic and visual cues are common nonverbal feedback used in virtual experiences. Although visual cues are self-descriptive, it is still challenging for users to understand haptic information due to

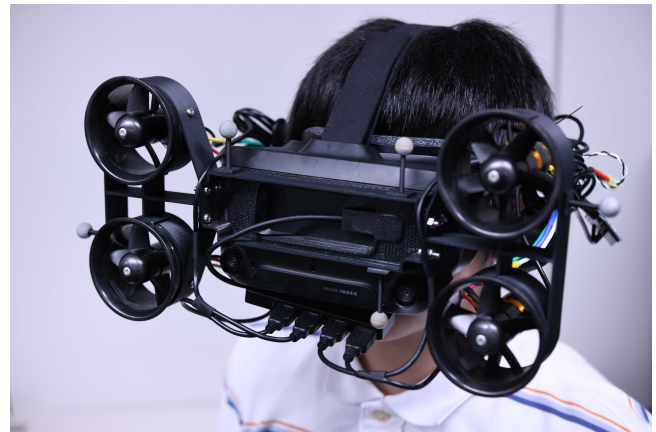


Figure 1: X-Wing system overview.

limited force feedback technology developed to date. For instance, haptic feedback through direct stimulation such as vibration [Chang et al. 2018], or indirect stimulation through the exertion of force [Liu et al. 2020] can lead to a different experience and use cases. Therefore, haptic feedback can enhance the experience and shrink gaps between direct and indirect stimulation.

Lately, haptic technology such as force feedback devices is made into wearable or stationary. A wearable force feedback device offers better flexibility in user movement yet, it is hard to induce intensive haptic feedback. On the other hand, a stationary device could produce a stronger force in exchange for portability. Therefore, it becomes a tough decision when a task requires both forceful and portable haptic feedback.

To overcome the challenge, we propose a portable force feedback device, offering a strong force power by using thrust force generated with ducted fans. Compared to a similar handheld technique [Sasaki et al. 2018] [Heo et al. 2018], our work uses a head-based design for a hand-less experience. It can also transform by applying the force on a user's head into a resultant force to alter the user's head direction similar to KABUTO [Tanichi et al. 2020].

In this demonstration, we introduce X-Wing, a wind-based force feedback system designed by attaching four ducted fans to an HMD, producing thrust force for a unique virtual experience. Our main contribution is that it explores the existing research gap of a portable

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force feedback device with an intensive force power (22.76N). Also, our system supports providing continuous thrust energy for force feedback to the user's head with translation and rotation force.

2 PROTOTYPE

Figure 1 illustrates the overview of the prototype attached to a Valve Index VR HMD. Four fans were made using a 64mm ducted airplane fan (QX-MOTOR QF2822) with 3D printed protection cases. An Arduino Nano was used to convert digital information based on VR user interaction in the Unity game engine in real-time and sent the serial data to an Electric Speed Controller (ESC). The ESC manipulated the respective fans' speed using the Pulse Width Modulation (PWM) signal. A lithium-ion battery (1200mAh, 11.1V, 20C) was also connected to the ESC as a portable power supply. In contrast, the system weighted 850g (4 Fans, Circuits and Batteries) and 1.6kg (HMD included). One fan generates 86dB at 2N in forward rotation, 92.6dB at the maximum force (5.68N) in forward rotation, and 110.6dB at the maximum force (2.35N) in reverse rotation. Users are suggested to wear earplugs in the demonstration as the noises will result in distractions to the user experience.

2.1 System configuration

The PWM signal allows the ducted fans to rotate in both clockwise and anticlockwise directions. Hence, it enables the system to produce thrust forces in six directions: Forward, Backward, Upward, Downward, Leftward and Rightward.

Figure 2 illustrates a different directional force generated by altering the rotating direction of ducted fans based on events of collisions occurred synchronously in the virtual environment. The maximum rotational speed of the pulse (μs) from the PWM is needed to calculate the force (N) exerted in each direction by the ducted fans. With the existing configuration, the capped clockwise rotation could provide a pulse of $1900\mu s$, producing 5.69N of force. On the other side, the rotation for anticlockwise is capped at $1100\mu s$, producing a 2.35N amount of forces. The horizontal and vertical distances of each ducted fan are 0.135m and 0.050m, respectively. With the data, the total thrust force in each direction can be calculated as follows:

Backward: $2.35N \times 4 = 9.4N$

Forward: $5.69N \times 4 = 22.76N$

Yaw: $5.69N \times 2 \times 0.135m + 2.35N \times 2 \times 0.135m = 2.17Nm$

Pitch: $5.69N \times 2 \times 0.050m + 2.35N \times 2 \times 0.050m = 0.80Nm$

Despite the force, the system has a latency of 200ms from sending signals by Unity to the ducted fans. In addition, it takes a further 100ms to 300ms for the ducted fans to operate at the peak speed. Hence, the system has a total latency of around 500ms.

3 DEMONSTRATION

By attaching ducted fans on an HMD and operating at a certain speed, a user can experience a unique virtual experience with the haptic feedback (e.g., experiencing inertial force and momentum when driving a vehicle). At the SIGGRAPH Asia 2021, we will demonstrate a virtual flight simulator where attendees will be using our system to experience haptic feedback in a virtual environment.

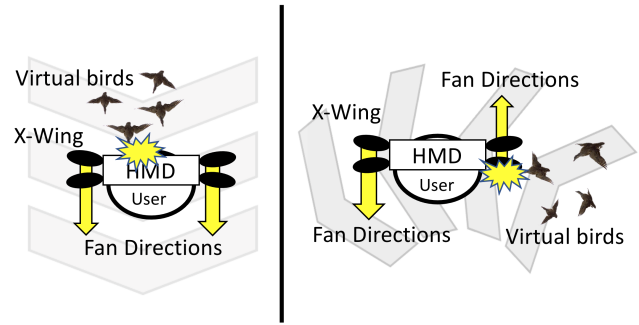


Figure 2: Top-down view showing ducted fans rotate in different combinations of directions to generate directional forces according to the types of collisions.

Attendees will experience different forces according to their movement and inertial force in the virtual environment. For example, it could be triggered by entering a cloud or colliding with objects such as birds during their virtual interaction.

4 CONCLUSION

In this demonstration, we introduce X-Wing: a propeller-based force feedback system attached to the user's head. This system explores the existing research gap by proposing a portable force feedback device that can create a comparable thrust force to a stationary setup. We used four ducted fans by attaching them to a VR HMD to produce thrust energy at different speeds and directions for immersive haptic stimulation. It allows a VR user to stimulate inertial force and realize momentum in a virtual environment for an enhanced user experience.

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REFERENCES

- Hong-Yu Chang, Wen-Jie Tseng, Chia-En Tsai, Hsin-Yu Chen, Roshan Lalintha Peiris, and Liwei Chan. 2018. FacePush: Introducing Normal Force on Face with Head-Mounted Displays. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (Berlin, Germany) (UIST '18). Association for Computing Machinery, New York, NY, USA, 927–935. <https://doi.org/10.1145/3242587.3242588>
- Seongkook Heo, Christina Chung, Geehyuk Lee, and Daniel Wigdor. 2018. Thor's Hammer: An Ungrounded Force Feedback Device Utilizing Propeller-Induced Propulsive Force. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3173574.3174099>
- Shi-Hong Liu, Pai-Chien Yen, Yi-Hsuan Mao, Yu-Hsin Lin, Erick Chandra, and Mike Y. Chen. 2020. HeadBlaster: A Wearable Approach to Simulating Motion Perception Using Head-Mounted Air Propulsion Jets. *ACM Trans. Graph.* 39, 4, Article 84 (July 2020), 12 pages. <https://doi.org/10.1145/3386569.3392482>
- Tomoya Sasaki, Richard Sahala Hartanto, Kao-Hua Liu, Keitarou Tsuchiya, Atsushi Hiyama, and Masahiko Inami. 2018. Leviopole: Mid-Air Haptic Interactions Using Multirotor. In *ACM SIGGRAPH 2018 Emerging Technologies* (Vancouver, British Columbia, Canada) (SIGGRAPH '18). Association for Computing Machinery, New York, NY, USA, Article 12, 2 pages. <https://doi.org/10.1145/3214907.3214913>
- Taku Tanichi, Futa Asada, Kento Matsuda, Danny Hynds, and Kouta Minamizawa. 2020. KABUTO: Inducing Upper-Body Movements Using a Head Mounted Haptic Display with Flywheels. In *SIGGRAPH Asia 2020 Emerging Technologies* (Virtual Event, Republic of Korea) (SA '20). Association for Computing Machinery, New York, NY, USA, Article 9, 2 pages. <https://doi.org/10.1145/3415255.3422880>